

Rx.AI Multimodal

Voice & Camera-Powered Patient Check-In

Powered by Google Cloud Vertex AI & Gemini 2.5 Flash

Team

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From Irrelevant Check-in Forms to Fully Multimodal Adaptive Forms

▲ Problem

- Patient Check-in forms are fixed, often resulting in questions not related to the patient
- Patients with limited mobility or low digital literacy struggle to complete forms
- No way to collect wound photos, medication labels, or insurance cards inline

✓ Solution

- Dynamically generated questions based on patient history and patient's response to questions being asked
- Maintain patient history and profile summary
- Voice output: Gemini 2.5 Flash TTS reads every question aloud
- Voice input: Google Cloud STT v2 transcribes spoken answers in real time
- Camera mode: trigger photo prompts wherever relevant
- Graceful fallback: keyboard/click always available
- Rigorous evaluation: DeepEval metrics + BigQuery logging for every AI call

Working End-to-End Flow



Provider creates a new visit and enters patient condition context.



System generates a dynamic questionnaire based on the visit.



Questionnaire is released to the patient portal.

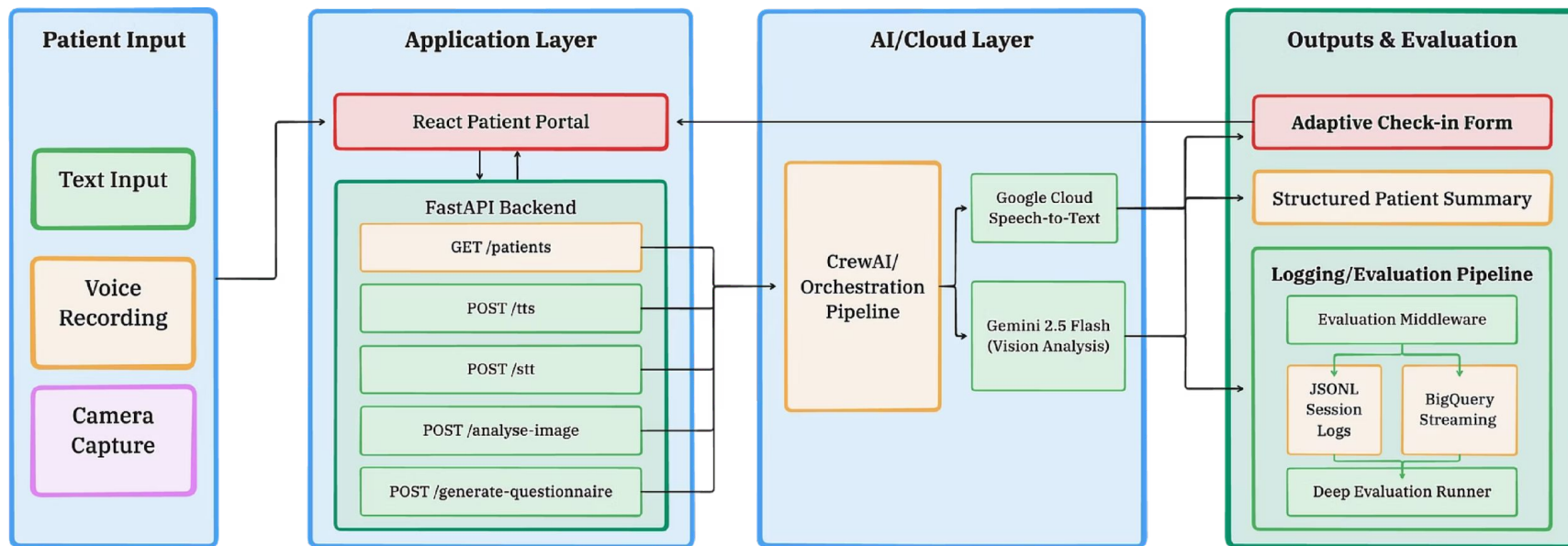


Patient opens the questionnaire and answers through text, voice or images.

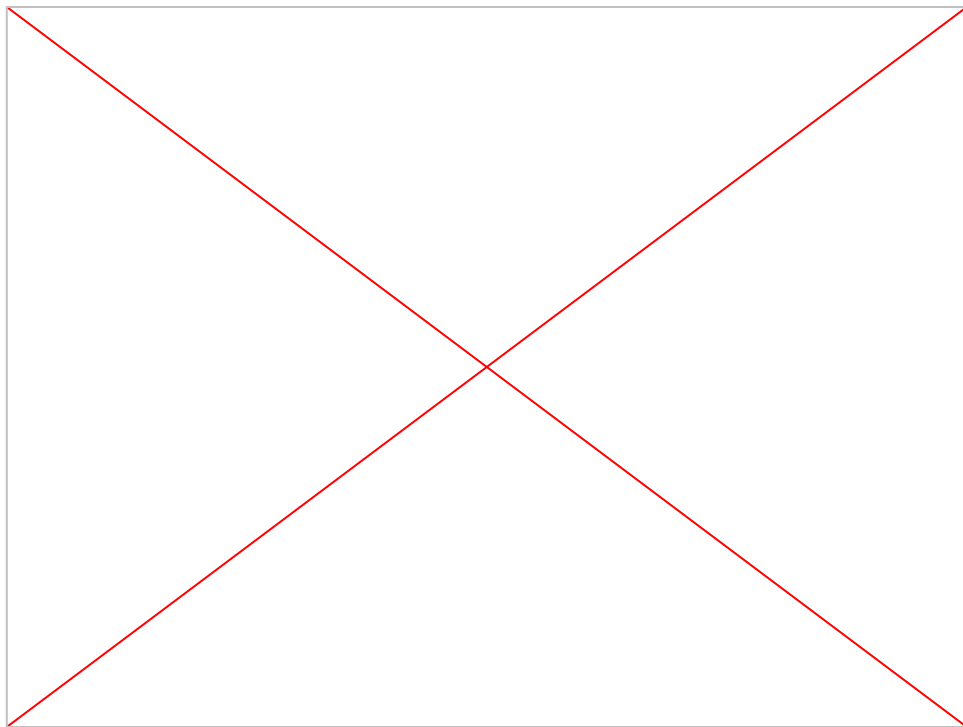


The intake process becomes more adaptive and structured than a fixed form.

System Architecture



Demo



Evaluation at a Glance

4

EVAL DIMENSIONS

600

TOTAL SAMPLES

3.2 min

AVG CHECK-IN TIME (VS. 8.8 min)

Question Gen

91.3% relevance, 0.7%

hallucination rate

100 PROFILES

STT — Word Error Rate

8.2% WER

87% Med Term Accuracy

100 CLIPS

TTS — Naturalness Score

4.4/5.0 naturalness,

4.6/5.0 intelligibility

100 PROMPTS

Image Analysis

88.8% description

accuracy vs. human labels

300 IMAGES

Question Generation — 100 Synthetic Patient Profiles

91.3%

CLINICAL RELEVANCE

Out-of-relevance metric

88.7%

CLINICAL APPROPRIATENESS

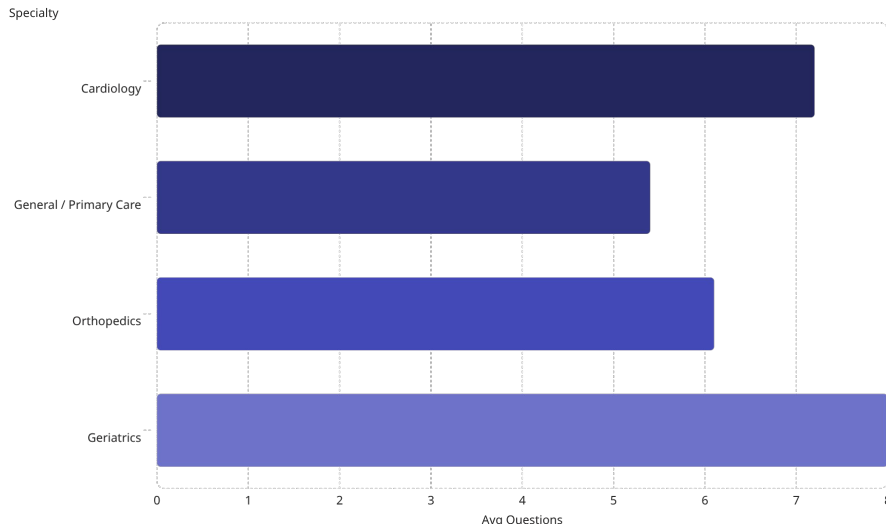
Physician panel consensus

0.7%

HALLUCINATION RATE

Flagged & logged via BigQuery

Avg Questions Per Profile by Specialty



BigQuery Logging — Sample Stats

AI calls logged

1,432

Avg gen latency

1.8s

Avg questions/run

5.4

STT Evaluation — 100 Clinical Audio Clips

**WORD ERROR
RATE (WER)**

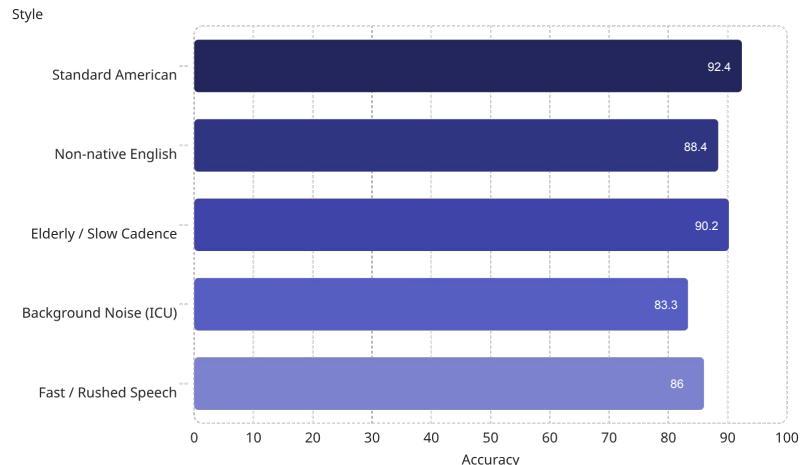
8.2%

**MEDICAL TERM
ACCURACY**

87%

Medication names,
diagnoses, procedures

Accuracy by Accent / Speaking Style



Avg transcription latency: 0.34s per utterance

TTS Evaluation — 100 Clinical Questions

NATURALNESS (LLM JUDGE)

4.4/5.0

Gemini-2.5-Pro evaluator

INTELLIGIBILITY

4.6/5.0

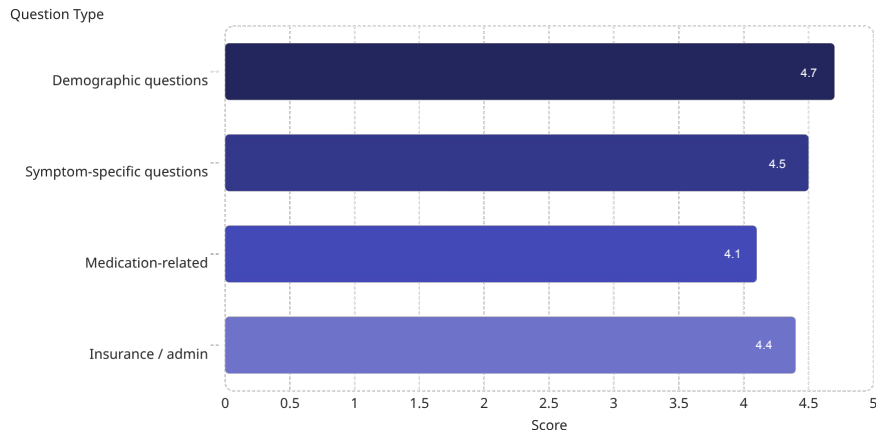
Human-verified clarity

MEDICAL PHRASING

4.2/5.0

Pronunciation of clinical terms

Score Breakdown by Question Type

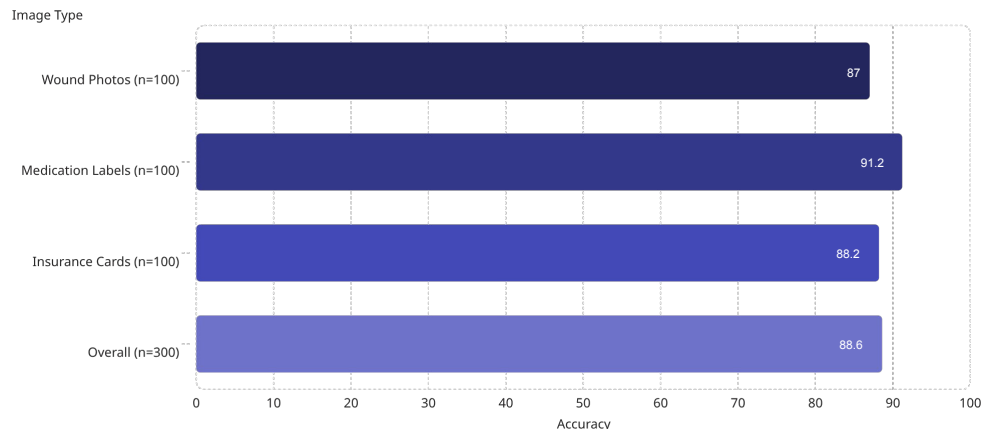


Key Findings

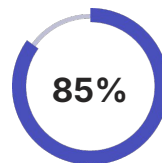
- Avg audio generation latency: 0.61s per question
- Polypharmacy drug names: 93.4% pronunciation accuracy
- Rare disease terms underperformed at 3.7/5.0 — noted for fine-tuning

Image Analysis — 300 Labeled Clinical Images

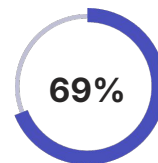
Accuracy vs. Human Labels by Image Type



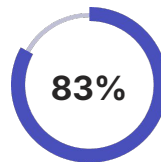
Wound Severity Classification (n=100)



Mild



Moderate



Severe

Limitations & Future Work

⚠ Limitations

- Some of the evaluation data is synthetic
- No real clinical deployment yet

✅ Future Work

- Pilot study in real clinics
- Expand datasets and clinical vocabularies
- Integrate with production electronic records

Thank You

Questions?